SAKSHAM AGGARWAL

AI Engineer & Data Science Undergraduate | Multi-Agent LLMs, RAG Pipelines, Computer Vision & ML

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PROFILE SUMMARY

AI Engineer and Data Science undergraduate (**CGPA 8.98**) with proven, production-grade experience building **multi-agent LLM systems, RAG pipelines, and end-to-end AI applications**. Deep expertise in Python, Scikit-learn, FAISS, LangGraph, OpenCV, SQL, and cloud platforms. **Winner of a national hackathon (1st of 600+ teams)**; certified across **Microsoft Azure AI, Data Science & Fabric Analytics** and Oracle Cloud Infrastructure. Passionate about building intelligent systems that create real, measurable impact.

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU) Expected 2027

Bachelor of Technology (BTech) in Artificial Intelligence & Data Science | **CGPA: 8.98 / 10.0**

St. Michael's Senior Secondary School — Class XII (CBSE): 76.2% (2023) | Class X (CBSE): 96.8% (2021)

TECHNICAL SKILLS

LLM & Multi-Agent AI: LangGraph, LangChain, RAG Architecture (FAISS, Sentence-Transformers), Natural Language Processing (NLP), Text Embeddings & Tokenization, Multi-Agent Orchestration, Groq, Claude / OpenAI / GPT4All APIs, Prompt Engineering

ML & Data Science: Scikit-learn (Classification, Regression, Cross-Validation), Statistical Analysis, Pandas, NumPy, Model Evaluation, OpenCV, MediaPipe (Computer Vision), TensorFlow / PyTorch (Foundations)

Programming & Engineering: Python (OOP, Async, Scripting), SQL (PostgreSQL, MySQL), FastAPI, Streamlit, REST APIs, Docker, Pytest (Automated Testing), Git/GitHub, Linux/CLI, Excel Automation (OpenPyXL)

Cloud & Certifications: Microsoft Azure (AI Engineer AI-102, Data Scientist DP-100, Fabric Analytics DP-600 — All Certified), Oracle Cloud Infrastructure (OCI), GCP Fundamentals, Azure Synapse, Power BI

PROJECTS

Adaptive Multi-Agent & RAG Knowledge Management System GitHub ↗ Python · LangGraph · FAISS · Scikit-learn · FastAPI · Streamlit · Docker

- Engineered a production **multi-agent LLM orchestration system** using **LangGraph** — Resume Parser, Question, Evaluation, and Feedback agents — conducting fully adaptive technical interviews grounded in a live FAISS RAG knowledge base.
- Built a high-precision **FAISS vector retrieval pipeline** achieving **100% Top-1 topic retrieval, 96.7% Recall@3, and a perfect 1.0 MRR**; trained a **Scikit-learn classifier at 97.92% cross-validated accuracy** to score candidate readiness.
- Shipped as a containerized full-stack application (FastAPI + Streamlit) with a **25-test automated Pytest suite** ensuring continuous pipeline integrity — end-to-end ownership from data ingestion to live deployment.

AI Vision Telemetry & Local LLM Agent with Tool-Calling GitHub ↗ Python · OpenCV · MediaPipe · GPT4All · Notion API · OCR

- Designed a real-time computer vision pipeline using **MediaPipe pose landmarks** achieving **~90% tracking accuracy** across 10+ exercises; delivered live visual feedback and automated rep counting improving form accuracy by **~25–30%**.
- Integrated an on-premise **Local LLM (GPT4All)** with RAG-style OCR document retrieval to ground model responses in uploaded physiotherapy documents without cloud API dependency.
- Built a **tool-calling integration to Notion's database API** enabling the assistant to autonomously log, query, and summarize session history — direct production experience connecting LLMs to live application backends.

Crypto Analyst Pro — Real-Time Market Data & LLM Synthesis Engine Python · Anthropic Claude API · Scikit-learn · Prompt Engineering · REST APIs

- Built an automated pipeline extracting real-time price feeds, volume trends, and news streams via financial REST APIs; designed structured **Claude API** prompt pipelines converting raw market data into quantifiable sentiment and risk scores.
- Trained **Scikit-learn predictive models** through feature engineering, cross-validation, and statistical backtesting — boosting directional forecast accuracy by **~60–75%**.

PROFESSIONAL EXPERIENCE & CERTIFICATIONS

Microsoft Learn Student Ambassador Jun 2024 – Present

- Conducted technical workshops on **Azure AI services, cloud data analytics, and modern Python engineering** for 200+ students and developers; represented Microsoft's AI and cloud ecosystem at campus and community events.
- **Industry Certifications:** Azure AI Engineer Associate (AI-102), Azure Data Scientist Associate (DP-100), Microsoft Fabric Analytics Engineer Associate (DP-600); Oracle Cloud Infrastructure (OCI) Foundations Associate.

Open-Source Contributor — GirlScript Summer of Code (GSSoC) Oct 2024 – Nov 2024

- Contributed to production Python codebases — feature development, debugging, documentation, and API testing. Certified as a **Postman API Student Expert**.

KEY ACHIEVEMENTS & HONORS

- **Winner (1st Place), Zero-to-One National Hackathon (Chandigarh University):** Ranked #1 out of 600+ competing engineering teams for architecting an autonomous, high-impact AI solution (featured: ANI News, MadeInMedia).
- **Runner-Up, IIIT Delhi HackLLM Hackathon:** Engineered and deployed a functional LLM-driven application within strict 24-hour constraints (LinkedIn Post).
- **National Hackathon Finalist:** Selected as finalist across 5+ national-tier AI, Machine Learning, and Computer Vision hackathons.